

Drug Interaction Dataset

24 drugs · 11 active metabolites · schema 1.5.0

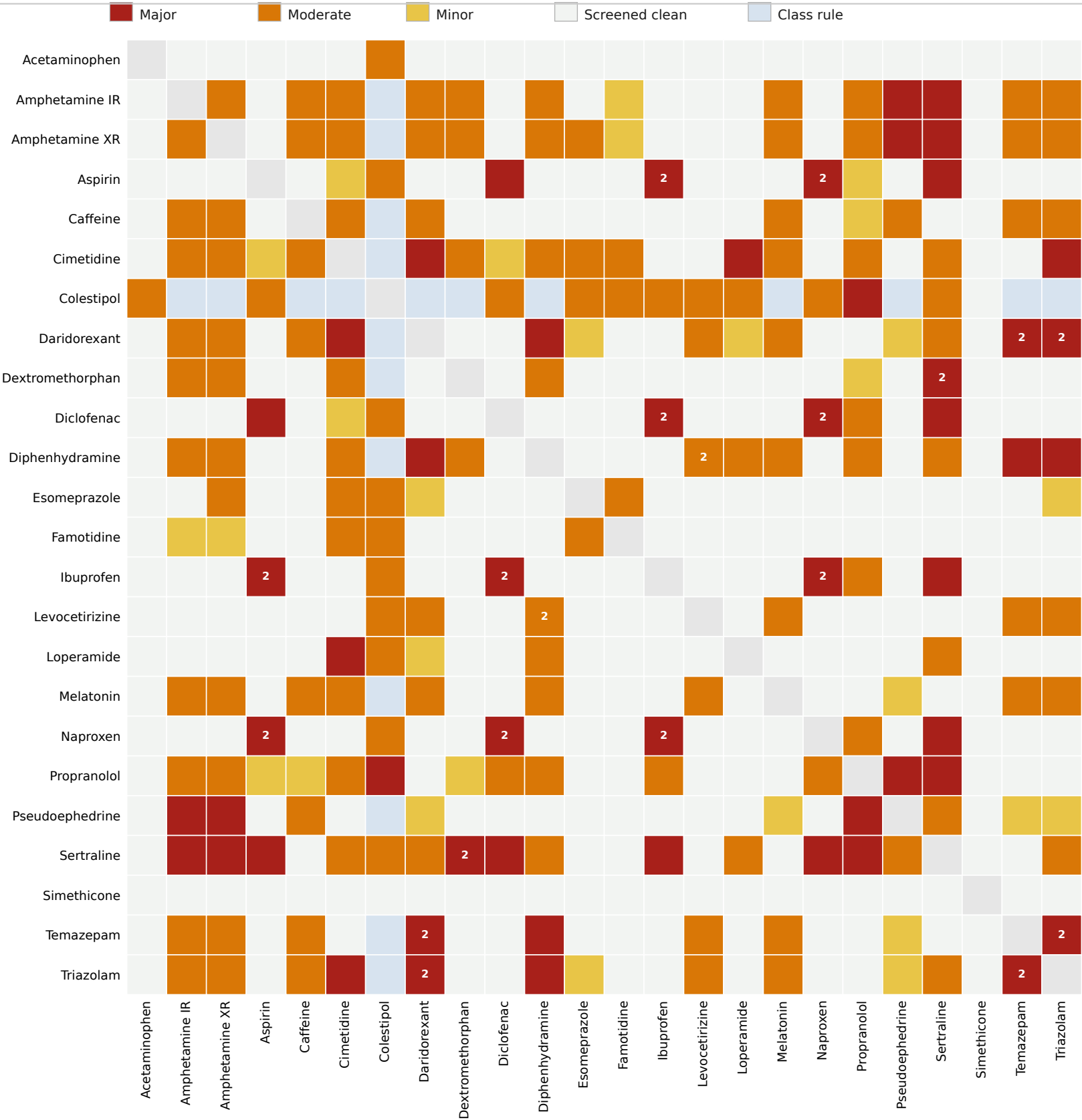
Drug records	24
Active metabolite records	11
Controlled vocabularies	13
Drug pairs (24 choose 2)	276
Interaction findings	116
major	36
moderate	66
minor	14
Explicit screening entries	171
Pair coverage	100%

Sources: Merck Manual Professional · AHFS DI (Drugs.com) · DailyMed
StatPearls (NCBI) · DrugBank

**Reference data assembled from tertiary sources.
Not a clinical decision tool. Not reviewed by a pharmacist.**

Interaction matrix

All 276 pairs. Colour is the highest severity found; a number marks pairs with more than one distinct finding.
Pale cells were screened and found clean — a recorded result, not an untested gap.



Symmetric. The stored records write each finding once, on whichever drug comes later in canonical order.

Major interactions

35 drug-drug findings graded major, grouped by mechanism. Severity grades the consequence; certainty is recorded separately as evidence. Suggested actions are in the records.

SEROTONERGIC AND CYP2D6 - THE SERTRALINE CLUSTER

Sertraline + Amphetamine IR Additive serotonergic activity, plus additive cardiovascular effect. Sertraline also weakly inhibits CYP2D6, a minor amphetamine route.	<i>additive toxicity</i>
Sertraline + Amphetamine XR Additive serotonergic activity, plus additive cardiovascular effect. Sertraline also weakly inhibits CYP2D6, a minor amphetamine route.	<i>additive toxicity</i>
Sertraline + Aspirin SSRIs deplete platelet serotonin and impair platelet aggregation; NSAIDs damage the mucosa and inhibit platelet COX-1. Combined, upper GI bleeding risk rises several-fold above either alone. One of the most under-recognised pairings in ordinary outpatient use.	<i>additive toxicity</i>
Sertraline + Dextromethorphan The clearest serotonin syndrome risk in this database, and it is a double hit: sertraline inhibits CYP2D6, which is dextromethorphan's main clearance route, so parent levels rise several-fold at the same time as serotonergic tone rises. Cases are well documented at ordinary OTC doses.	<i>additive toxicity</i>
Sertraline + Dextromethorphan CYP2D6 inhibition by sertraline converts an extensive metaboliser into a functional poor metaboliser. Recorded separately from the pharmacodynamic entry because the two need different mitigations: one is solved by dose, the other is not.	<i>metabolism</i>
Sertraline + Diclofenac SSRIs deplete platelet serotonin and impair platelet aggregation; NSAIDs damage the mucosa and inhibit platelet COX-1. Combined, upper GI bleeding risk rises several-fold above either alone. One of the most under-recognised pairings in ordinary outpatient use.	<i>additive toxicity</i>
Sertraline + Ibuprofen SSRIs deplete platelet serotonin and impair platelet aggregation; NSAIDs damage the mucosa and inhibit platelet COX-1. Combined, upper GI bleeding risk rises several-fold above either alone. One of the most under-recognised pairings in ordinary outpatient use.	<i>additive toxicity</i>
Sertraline + Naproxen SSRIs deplete platelet serotonin and impair platelet aggregation; NSAIDs damage the mucosa and inhibit platelet COX-1. Combined, upper GI bleeding risk rises several-fold above either alone. One of the most under-recognised pairings in ordinary outpatient use.	<i>additive toxicity</i>
Sertraline + Propranolol Moderate CYP2D6 inhibition by sertraline against a high-extraction CYP2D6 substrate. The effect scales with sertraline dose — slight at 50 mg, substantial at 200 mg — which means the interaction can appear weeks after both drugs were started.	<i>metabolism</i>

NSAID STACKING AND BLEEDING RISK

Diclofenac + Aspirin Additive gastrointestinal, renal and cardiovascular toxicity with no additive analgesic benefit — the COX enzyme is already saturated by one agent at therapeutic dose. GI bleeding risk rises multiplicatively.	<i>additive toxicity</i>
Ibuprofen + Aspirin Additive gastrointestinal, renal and cardiovascular toxicity with no additive analgesic benefit — the COX enzyme is already saturated by one agent at therapeutic dose. GI bleeding risk rises multiplicatively.	<i>additive toxicity</i>

Major interactions (continued)

Continued from the previous page.

Ibuprofen + Aspirin

antagonism

Ibuprofen occupies the COX-1 channel reversibly and blocks aspirin from reaching serine-529, so aspirin cannot acetylate it. The cardioprotective effect of low-dose aspirin is lost while this continues. This is the reason aspirin's irreversible mechanism is recorded as its own term rather than as generic COX inhibition.

Ibuprofen + Diclofenac

additive toxicity

Additive gastrointestinal, renal and cardiovascular toxicity with no additive analgesic benefit — the COX enzyme is already saturated by one agent at therapeutic dose. GI bleeding risk rises multiplicatively.

Ibuprofen + Diclofenac

duplication

Two COX inhibitors for one job.

Naproxen + Aspirin

additive toxicity

Additive gastrointestinal, renal and cardiovascular toxicity with no additive analgesic benefit — the COX enzyme is already saturated by one agent at therapeutic dose. GI bleeding risk rises multiplicatively.

Naproxen + Diclofenac

additive toxicity

Additive gastrointestinal, renal and cardiovascular toxicity with no additive analgesic benefit — the COX enzyme is already saturated by one agent at therapeutic dose. GI bleeding risk rises multiplicatively.

Naproxen + Diclofenac

duplication

Two COX inhibitors for one job.

Naproxen + Ibuprofen

additive toxicity

Additive gastrointestinal, renal and cardiovascular toxicity with no additive analgesic benefit — the COX enzyme is already saturated by one agent at therapeutic dose. GI bleeding risk rises multiplicatively.

Naproxen + Ibuprofen

duplication

Two COX inhibitors for one job.

SEDATIVE STACKING

Diphenhydramine + Daridorexant

additive CNS depression

Additive sedation, psychomotor impairment and next-day residual effect. The mechanisms differ (GABA-A, H1, orexin, MT1/MT2) so the effect is not detected by any duplication check that reasons from drug class alone.

Temazepam + Daridorexant

additive CNS depression

Additive sedation, psychomotor impairment and next-day residual effect. The mechanisms differ (GABA-A, H1, orexin, MT1/MT2) so the effect is not detected by any duplication check that reasons from drug class alone.

Temazepam + Daridorexant

duplication

Different mechanisms, same indication and same night.

Temazepam + Diphenhydramine

additive CNS depression

Additive sedation, psychomotor impairment and next-day residual effect. The mechanisms differ (GABA-A, H1, orexin, MT1/MT2) so the effect is not detected by any duplication check that reasons from drug class alone.

Triazolam + Daridorexant

additive CNS depression

Additive sedation, psychomotor impairment and next-day residual effect. The mechanisms differ (GABA-A, H1, orexin, MT1/MT2) so the effect is not detected by any duplication check that reasons from drug class alone.

Triazolam + Daridorexant

duplication

Different mechanisms, same indication and same night.

Triazolam + Diphenhydramine

additive CNS depression

Additive sedation, psychomotor impairment and next-day residual effect. The mechanisms differ (GABA-A, H1, orexin, MT1/MT2) so the effect is not detected by any duplication check that reasons from drug class alone.

Major interactions (continued)

Continued from the previous page.

Triazolam + Temazepam

additive CNS depression

Additive sedation, psychomotor impairment and next-day residual effect. The mechanisms differ (GABA-A, H1, orexin, MT1/MT2) so the effect is not detected by any duplication check that reasons from drug class alone.

Triazolam + Temazepam

duplication

Same class, same receptor site, same indication. Also recorded as additive CNS depression above; both entries are intentional, because a duplication check and a sedation check should each catch this.

CYP3A4 INHIBITION

Daridorexant + Cimetidine

metabolism

89% of daridorexant's metabolic clearance is CYP3A4, so its exposure is close to a direct function of 3A4 activity. Next-morning driving impairment is the dose-related risk the label caps for.

Loperamide + Cimetidine

metabolism

Loperamide's safety rests on almost none of it reaching the systemic circulation. Cimetidine inhibits CYP3A4 — one of its two clearance routes — raising systemic exposure toward the range where hERG blockade prolongs QT. The FDA boxed warning for loperamide is about exactly this scenario.

Triazolam + Cimetidine

metabolism

Triazolam is cleared almost entirely by CYP3A4 and has a sub-milligram therapeutic dose, so it has essentially no margin. Cimetidine inhibits CYP3A4. Expect prolonged sedation, next-day impairment and anterograde amnesia.

SYMPATHOMIMETIC AND UNOPPOSED ALPHA

Pseudoephedrine + Amphetamine IR

synergism

Additive sympathomimetic effect. Amphetamine and pseudoephedrine share the indirect noradrenaline-releasing mechanism; caffeine adds through adenosine antagonism. None of these is detected by a same-class duplication check.

Pseudoephedrine + Amphetamine XR

synergism

Additive sympathomimetic effect. Amphetamine and pseudoephedrine share the indirect noradrenaline-releasing mechanism; caffeine adds through adenosine antagonism. None of these is detected by a same-class duplication check.

Pseudoephedrine + Propranolol

antagonism

The textbook unopposed-alpha interaction: pseudoephedrine drives alpha-1 vasoconstriction while propranolol blocks the beta-2 vasodilation that would normally offset it. Hypertensive episodes with reflex bradycardia are documented, including in people whose blood pressure is otherwise well controlled.

ABSORPTION

Propranolol + Colestipol

absorption

Specifically documented on the colestipol label: repeated colestipol doses before a propranolol dose reduced propranolol absorption. The reverse risk matters too — stopping colestipol without adjusting propranolol raises propranolol exposure.

Metabolic routes and enzyme effects

S = substrate (cleared by it) · I = inhibitor (raises other substrates) · S/I = both.
Every pharmacokinetic interaction in the dataset is an I in one row meeting an S in the same column.

	1A2	2B6	2C8	2C9	2C19	2D6	2E1	3A4	Glucuronidation	Hydrolysis	Renal, unchanged	Unabsorbed
Acetaminophen							S		S	S		
Amphetamine IR						S					S	
Amphetamine XR						S					S	
Aspirin									S	S		
Caffeine	S											
Cimetidine	I				I	I		I			S	
Colestipol												S
Daridorexant								S				
Dextromethorphan						S		S				
Diclofenac			S	S				S	S			
Diphenhydramine	S			S		S/I						
Esomeprazole					S/I			S				
Famotidine											S	
Ibuprofen			S	S					S			
Levocetirizine											S	
Loperamide		S	S			S		S				
Melatonin	S				S							
Naproxen	S			S					S			
Propranolol	S				S	S			S			
Pseudoephedrine											S	
Sertraline		S		S/I	S/I	S/I		S/I				
Simethicone												S
Temazepam								S	S			
Triazolam								S				

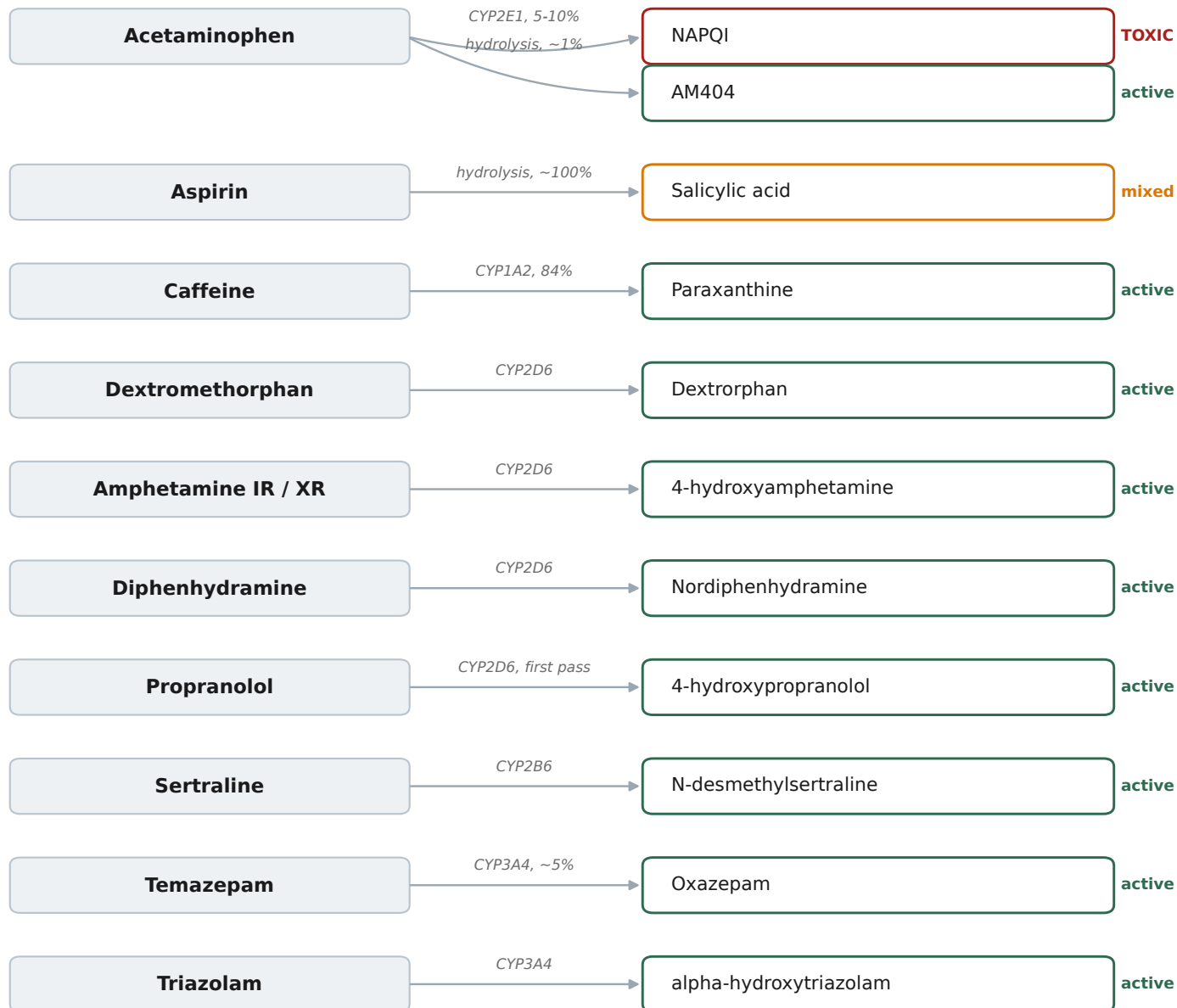
Read the two contrast pairs off this grid. Cimetidine carries four I marks; famotidine carries none — same drug class, same indication. Triazolam is a single S under 3A4 with no alternative route; temazepam sits under glucuronidation, which no inhibitor in this set touches. That is why cimetidine + triazolam is major and cimetidine + temazepam is nothing at all.

cytochrome P450

phase II and non-metabolic

Active metabolites

Metabolites get their own record when something is true of them that is not true of the parent: their own mechanism, their own disposal route, their own half-life, or their own interactions.



WHY THESE HAVE THEIR OWN RECORDS

NAPQI

Its own mechanism (protein arylation) and its own antidote. Too reactive to have plasma PK at all — recorded as a settled absence, not a pending one.

Salicylic acid

Carries almost all of aspirin's systemic activity. Saturable elimination, so its half-life runs 2-3 h at low dose and 15-30 h at high dose.

Dextrorphan

A CYP2D6 inhibitor raises the parent and suppresses this at the same time. Effects redistribute between two different receptor profiles rather than adding up.

N-desmethylsertraline

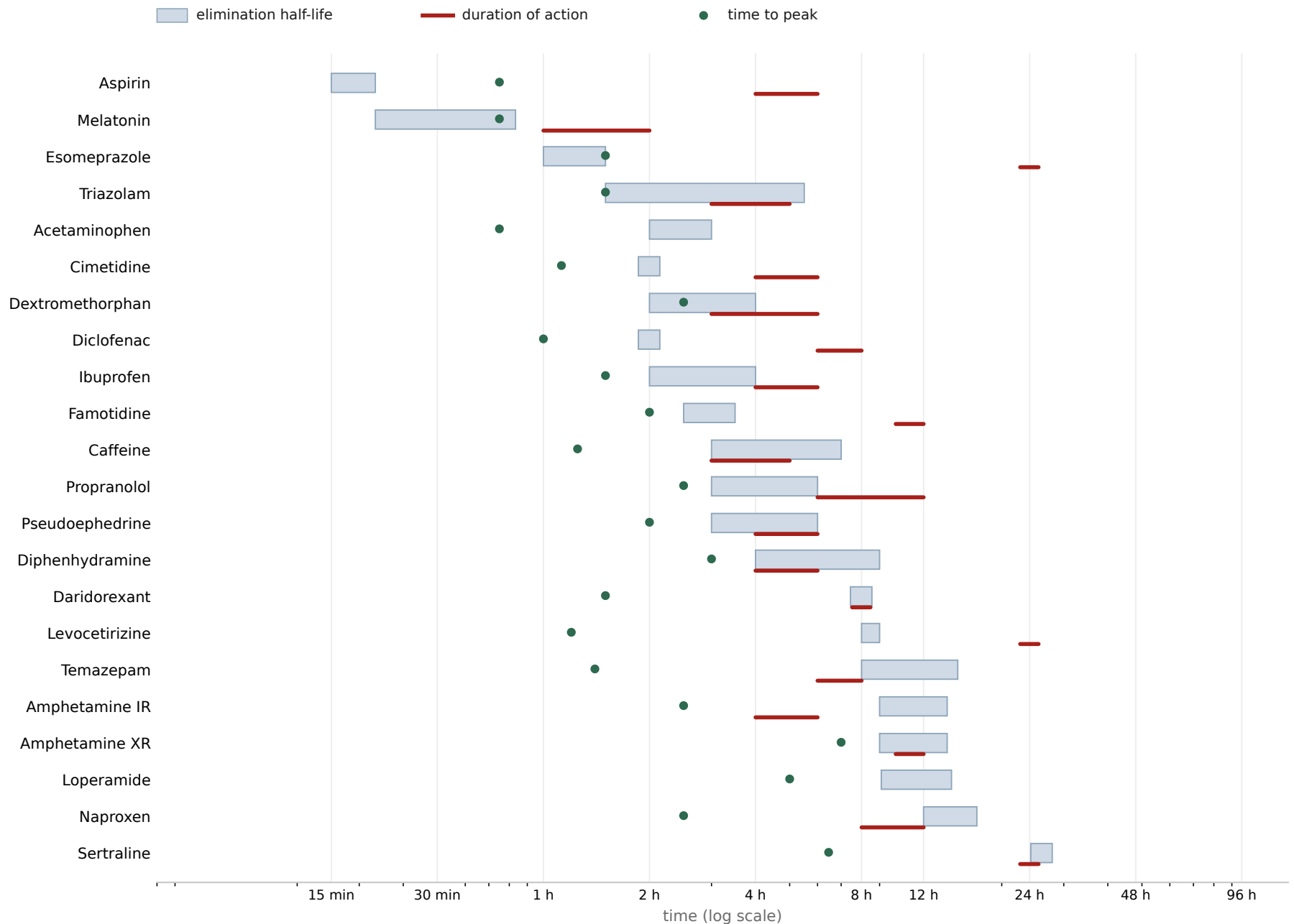
Only 1/20 as potent, but a 62-104 h half-life against the parent's 26 h. It governs the washout, which is why sertraline interactions outlast the last dose.

Paraxanthine

84% of a caffeine dose passes through it at comparable potency. Much of what is experienced as caffeine is really this.

Pharmacokinetics at a glance

Elimination half-life (bar) against duration of clinical effect (line) and time to peak (dot). Log scale.
Where the two disagree, the mechanism rather than the kinetics is setting the dosing interval.



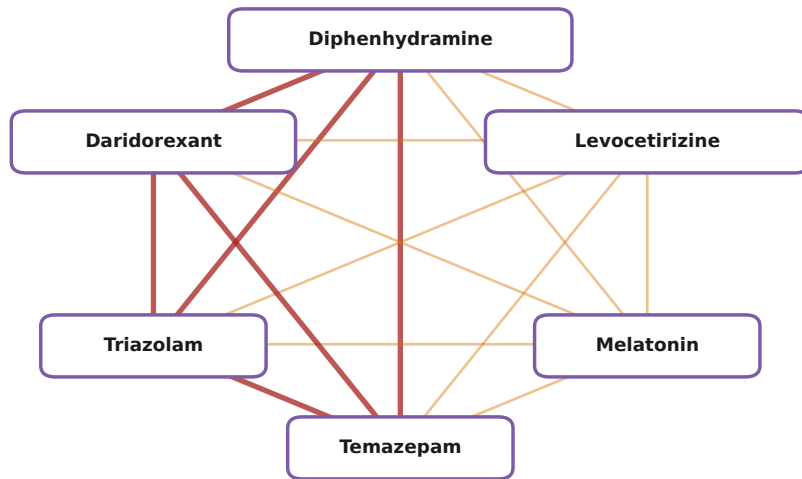
Aspirin is the clearest case where half-life misleads: 15-20 minutes for intact drug, 4-6 hours of analgesia carried by salicylate, and 7-10 days of antiplatelet effect set by platelet turnover. Esomeprazole is the same story — a 1 h half-life, 24 h of acid suppression, because it binds the pump covalently. Neither is derivable from the kinetics.

The two interaction hubs

Where findings concentrate. Neither cluster is caught by a duplication check that reasons from drug class.

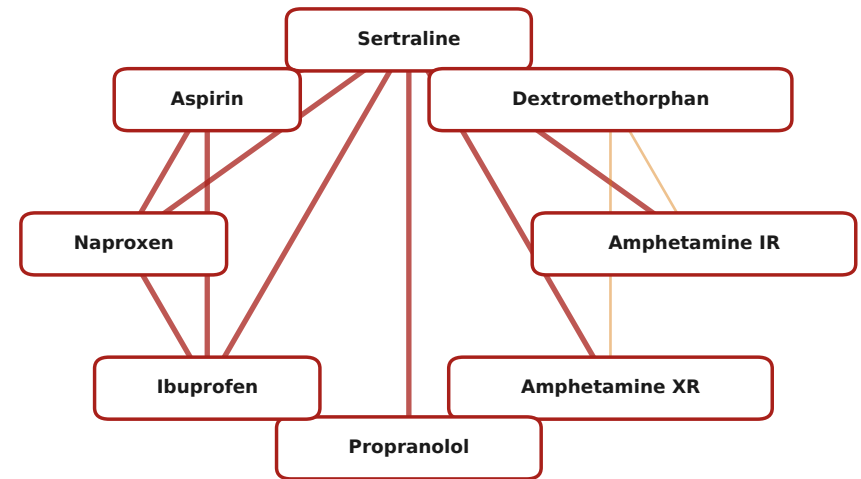
Sedative stacking

15 pairs, 6 of them major



Sertraline as hub

major against 7 of the other 23 records



Five sedatives, four mechanisms

GABA-A (temazepam, triazolam), H1 (diphenhydramine, levocetirizine), orexin (daridorexant) and MT1/MT2 (melatonin). Because the mechanisms differ, no class-based duplication check flags any of these fifteen pairs — but the impairment is additive regardless of mechanism.

One property, most of the findings

Moderate CYP2D6 inhibition explains sertraline against dextromethorphan and propranolol; platelet serotonin depletion explains it against all four NSAIDs. Both scale with sertraline dose, so the interaction can surface weeks after both drugs were started without anything changing.

What the data says to do

Acetaminophen is screened clean against sertraline and against every NSAID — the substitution that resolves the bleeding cluster. Famotidine is clean where cimetidine is major, and temazepam clean where triazolam is major. The negatives carry as much of the answer as the positives.